ME143 BASICS OF CIVIL & MECHANICAL ENGINEERING	
CO1	Introduce the important aspects and applications of mechanical engineering
CO2	Explain the working of different mechanical systems.
CO3	Understand the scope and basic elements of civil engineering.
CO4	Understand the concepts of building planning, surveying and properties of different building materials.

List of Experiment (ME143 Basics Of Civil & Mechanical Engineering)		
Sr. No.	**Title**	**Course Outcomes**
1	Introduction to Steam Boilers & Study of Cochran and Babcock-Wilcox Boilers	CO1
2	To Study Boiler Mountings & Accessories	CO1
3	Study of Vapour Compression Refrigeration Cycle & Windows Air-Conditioning System	CO1, CO2
4	Study of Power & Motion Transmission Systems	CO2
5	Study of Coupling, Clutches & Brakes	CO1, CO2
6	Introduction to surveying and basic surveying instruments	CO3
7	Linear measurement-chaining, ranging & offsetting	CO4
8	Construction site visit	CO4
9	Planning and drawing of residential building	CO3, CO4
10	Angular measurement-compass survey	CO4
11	Vertical measurement-dumpy level	CO4

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Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO. 1

INTRODUCTION TO STEAM BOILERS AND STUDY OF COCHRAN AND BABCOCK-WILCOX BOILER

AIM:
To introduce the types of steam boilers and study of Cochran Boiler and Babcock-Wilcox Boiler.

OBJECTIVE:
To study different types and classification of steam boilers and study of working and constructional features of Cochran Boiler and Lancashire Boiler.

THEORY:

BOILER

It is a closed vessel in which the heat produced by the combustion of fuel is transferred to water for its conversion into steam at the desired temperature and pressure. Broadly speaking, boiler is a device used for generating,

- Steam for power generation
- Hot water for heating purpose

Classification of Boilers:
The boilers may be classified as under:

According to the direction of the axis of the boiler shell:

1) **Horizontal Boiler:** If the axis of the boiler is horizontal, it is called the horizontal boiler.
   Example: Lancashire boiler, Locomotive boiler.
2) **Vertical Boiler:** If the axis of the boiler is vertical, it is called the vertical boiler. **Example:** Cochran boiler.
3) **Inclined Boiler:** If the axis of the boiler is inclined, it is known as inclined boiler.
   Example: Babcock & Wilcox boiler.

According to the tube content:

4) **Fire Tube Boiler:** The boilers in which the hot gases are inside the tubes and the water is surrounding the tubes is called fire tube boiler.
   Example: Lancashire boiler, Locomotive boiler, Cochran boiler.
5) **Water Tube Boiler:** The boiler in which the water is inside the tubes and the hot gases surround them is called water tube boilers.
   Example: Babcock & Wilcox boiler, Sterling boiler.

According to the furnace position:

1) **Externally Fired Boiler:** In the boiler if the fire is outside the shell, that boiler is known as externally fired boiler.
   Example: Babcock & Wilcox boiler.
2) **Internally Fired Boiler:** In the boiler in which the furnace is located inside the boiler shell, it is known as internally fired boiler.
   Example: Cochran boiler, Lancashire Boiler.

According to the circulation of water:

3) **Forced Circulation Boiler:** In the boilers if the circulation of water is done by a pump, then they are known as forced circulation boilers.
   Example: Velcox boiler, La-Mont boiler.
4) **Natural Circulation Boiler:** In the boiler if the circulation of water takes place due to difference in density resulting from difference in temperature, it is known as natural circulation boiler.
   Example: Babcock & Wilcox boiler, Lancashire boiler.

According to the working pressure:

5) **High pressure Steam Boiler:** If the working pressure of the boiler is more than 25 bar, the boiler is known as high-pressure steam boiler.
   Example: Babcock & Wilcox boiler.
6) **Medium Pressure Steam Boiler:** If the working pressure of the boiler is between 10 to 25 bar, the boiler is known as medium pressure steam boiler.
   Example: Lancashire boiler, Locomotive boiler.

3) **Low Pressure Steam Boiler:** If the working pressure of the boiler is between 3.5 to 10 bar, the boiler is known as low-pressure steam boiler.
   Example: Cochran boiler, Cornish boiler.

According to the use:

1) **Stationary Boilers:** The boilers, which cannot be transported easily from one place to another, are called stationary boilers.
   Example: Lancashire boiler, Babcock & Wilcox boiler.
2) **Portable boilers:** The boilers which can be easily transported (moved) from one place to another are called portable boilers.
   Example: Locomotive boiler.

According to the number of tubes:

1) **Single Tube Boiler:** The boiler having only one fire tube or water tube is called single tube boiler.
   Example: Cornish boiler.
2) **Multi Tube Boiler:** The boiler having two or more fire or water tubes for the circulation of hot gases or water are called multi tube boiler.
   Example: Lancashire boiler, Locomotive boiler, Babcock & Wilcox boiler, Cochran boiler.

A. COCHRAN BOILER

Specifications:

- Shell Dia =2.75 m
- Height = 5.75 m
- Working pressure=6.5 bar (max=15bar)
- Steam capacity=3500 kg/hr (max=4000kg)
- Heating section area=120 m^2
- Efficiency=70-75%

Characteristics:

- Vertical
- Portable
- Fire-tube
- Multi-tube
- Internally fired
- Natural circulation
- Solid as well as liquid fuel can be burnt

Working:

- The water is fed into boiler through feed check valve.
- Level is adjusted with the help of water level indicator.
- Coal is added trough a fire hole to grate and then burnt.
- The hot gases produced are collected in firebox.
- From fire box, the gases are entered into the combustion chamber with great velocity from the short fuel pipe or flue pix
- The firebrick lining in the combustion chamber deflects the hot gases to pass through horizontal tubes.
- From here the hot gases entered into the smoke box and exhaust to atmosphere through chimney.
- Water is evaporated and steam is collected at top of the boiler shell.
- Steam is then taken out through steam stop valve.

B. BABCOCK & WILCOX BOILER

Specifications:

- Diameter of the drum = 2000 to 4000 mm
- Length = 6000 to 9000 mm
- Size of water tubes = 76.2 to 101.6 mm
- Size of upper header tubes = 38.4 to 57.1 mm
- Maximum working pressure = 42 bar
- Maximum steaming capacity = 40,000 kg/hr.
- Efficiency = 60 to 80%

<u>Characteristics</u>:

- Horizontal
- Stationary
- Water – tube type
- Multi-tube boiler
- Externally fired
- Having natural circulation of water
- Having forced- circulation of air & hot gases
- Solid as well as liquid fuel fired
- High pressure

<u>Working</u>:

- Water is fed into shell through feed check value.
- Due to gravity water passes through the vertical tubes, headers & fills up the inclined tube first. The water then collects in boiler shell.
- Water level is adjusted with the help of water level indicator.
- Coal is added to trolley grate.
- Coal is then burnt.
- The trolley is then pushed into the fire space.
- In order to provide air for combustion of coal & circulation of hot gases, damper is lowered to the furnace level.
- The baffle plates do not allow hot gases to pass to the right side.
- These gases heat the super heater & then move down. These gases travel in the sine wave form and then move to the smoke-box fitted with a long chimney.
- Water in the upper portion of the tubes is heated and rises up. The cold water from the down take header rises to take its place & water from the boiler drum flows down. Thus natural circulation of water takes place by keeping the water tubes inclined.
- Till the temperature of water in the tubes & in the drum becomes uniform, the circulation of water continues. Finally the water in the upper portion of the tubes gets converted into steam.
- It is then collected at the top of the boiler shell.
- The steam collected above the surface of the water in the drum is wet steam. The super heater is used to convert this wet steam into superheated steam.
- The superheated steam is then taken out through the steam stop valve fitted on the top of the boiler for different uses.

QUESTIONS:

1. Define boiler & give classification of boilers.
2. State the applications of steam boilers.
3. Give the merits & demerits of fire tube boiler & water tube boiler.
4. Explain the working of Cochran boiler with neat sketch.
5. Explain the working of Babcock & Wilcox boiler with neat sketch.
6. State the applications & advantages of Babcock & Wilcox boiler.
7. State the function of following:
   Baffle plate
   Man hole
   Soot doors
8. How natural circulation is achieved in boiler by arrangement of tubes?

Conclusion:

Marks obtained:	Signature of faculty:	Date:

EXPERIMENT NO. 2
TO STUDY BOILER MOUNTINGS AND ACCESSORIES

AIM:
To study different types of boiler Mountings and Accessories.

OBJECTIVE:
To study the function, working and constructional features of boiler mountings and accessories.

THEORY:

A. <u>BOILER MOUNTINGS</u>

The fittings and devices, which are necessary for the safety and control of a boiler, are called Mountings.

According to I.B.R., the following mountings should be fitted to the boilers:
1. Two safety valves.
2. Two water level indicators.
3. A pressure gauge.
4. A steam stop valve.
5. A feed check valve.
6. A blow off cock.
7. A manhole.
8. Mud holes or sight holes.
9. Fusible plug.

Water Gauge

It indicates water level in the boiler. It consists of a glass tube, two gunmetal tubes & three cocks. The ends of the glass tube that passes through the stuffing boxes are connected with two metal tubes having flanges at their ends for bolting to the boiler. The water of the boiler comes into the glass tube through the lower tube and the steam through the upper tube. The water then stands in the glass tube at the same level as in the boiler. Two cocks are used to control the passage between the boiler and the glass tube while the third cock is in used to discharge some of the water from inside the boiler to see whether the gauge is in proper order or not. The glass tube is protected by means of a cover, made of specially toughened glass, which will prevent any accident that may happen due to breaking of glass tube. It is used for ordinary boilers.

Pressure Gauge

Function: The function of the pressure gauge is to measure the pressure exerted inside the vessel.

Construction: This gauge is generally fitted on the front side of the boiler shell. It is fitted to the steam space with the help of an inverted siphon.

It consists of a Bourdon's spring tube ABC. It is made of copper and bent into circular shape. This tube has an elliptical section. The end A of the tube is plugged. The other end C is connected to the hollow block E. The block E has a small opening for the entrance of steam or water from the boiler. The end A is connected to the link R. This is pivoted at H. This carries a toothed sector. This can swing about H. The small pinion meshes with this sector. Pinion carries a pointer P fitted on its spindle. To indicate pressure of steam inside the boiler, this pointer moves on a graduated dial or scale. The pressure gauge components are enclosed in a circular dial casing.

Working: The pressure gauge is connected with the boiler's steam space through a U-tube siphon. The U-shaped pipe contains water, which fills the Bourdon tube. The pressure of the steam acting through the water on the inside of the tube tries to make the tube circular. One end is fixed, the free end moves. Because of pinion & gear arrangement, the slight movement of the Bourdon tube is magnified considerably and the point gives a maximum deflection, which can be read on the scale.

Steam stop valve

Function: The function of the steam stop valve is to regulate the flow of steam from one steam pipe to the other or from the boiler to the steam pipe. The steam stop valve when directly mounted on the steam space of the boiler shell & connected to the steam pipeline which supplies steam to the prime mover is called a junction valve.

Working: When the hand wheel is turned in anticlockwise direction, the spindle is raised up. This will raise the valve from its seat. Thus a passage for the steam from the clearance between the valve and the valve seat is formed. In order to lower the valve, the hand wheel is rotated in clockwise direction. This rotation will close the passage for steam. Adjusting the position of the valve based on the requirements can regulate this. Under the normal working condition, the valve is open and the steam flows from the boiler to the steam pipe.

Feed check valve

Function: It is used to control the supply of water to the boiler & to prevent the escaping of water from the boiler when the pump pressure is less or pump is stopped.

Working: Under normal working condition, the pressure on the feed pump side (connected to elbow) is more than the boiler side pressure. This pressure difference lifts the check valve. To allow the feed water to enter the boiler, the feed valve is lifted manually. Hence, the feed water may enter the boiler. In order to control the supply of feed water to the boiler, the position of the feed valve is controlled. In the event of failure of feed pump, the pressure on the water sump side reduces. The check valve will be closed because of higher steam pressure. This will prevent the back flow of water from the boiler to the water sump. The check valve is to be replaced if it does not give satisfactory results. The stuffing box is provided in the valve to stop the leakage at the spindle.

Blow off cock valve

Functions:

- To discharge a portion of water when the boiler is in operation to blowout mud, scale or sediment periodically.
- To empty the boiler, when necessary for cleaning, inspection & repair.

It is fitted on the boiler shell directly or to a short branch at the lowest part of the water space. This pipe is known as blow-down pipe.

Working: In order to operate the valve, the rectangular slot is brought in line with the passage of the body. This is possible by rotating the plug with the help of wheel. When the slot is placed in this position, the cock is opened and all the impurities, mud, sediments etc. start flowing out of the boiler and they are removed. When the slot is brought at right angles to the passage of the body, the cock is closed.

Fusible Plug

Function: Its function is to protect the boiler against damage due to over heating for low water level. It is fitted in the crown of the furnace or firebox at appropriate place.
Construction: It consists of a hollow gunmetal body. This body is screwed into the firebox crown. The body has a hexagonal flange. This is used to tighten the body into the shell. A gunmetal plug having a hexagonal flange is screwed into the gunmetal body. The hollow gunmetal plug is fitted in another gunmetal plug. These two plugs are separated by an annulus of fusible metal.

Working: In the normal working conditions of the boiler, the fusible plug is fully submerged under water. This is so because under normal condition, proper water level is maintained & can be checked by water level indicator. Under the circumstances, the heat from the fusible plug is being conducted to water. This keeps the fusible metal at an almost constant temperature, much below its melting point. When the water level falls below the fusible plug, the plug gets uncovered from water. The upper portion of the plug gets exposed to the steam space. The steam cannot keep the plug cool. This will overheat the fusible metal. The plug falls down along with the fusible metal making a hole. The steam and water, being under pressure immediately reach the firebox and extinguish the fire.

Safety valves

Function: It is a device, attached to the steam space of the boiler shell, which opens automatically to discharge some steam & prevent the steam pressure inside the boiler to exceed.
There are **four types** of safety valves generally fitted on a boiler. Dead weight safety valve, Lever loaded safety valve, Spring-loaded safety valve, High steam & low water safety valve.

(1) Dead weight safety valve

A valve V is placed upon a valve seat S that is fixed upon a long vertical pipe P having a flange at the bottom for fixing at the top of the boiler. Suspended at the top of the valve is the weight carrier D that carries cast iron rings W. The total weight must be sufficient to keep the valve on its seat against the normal working pressure. When the steam pressure exceeds the normal limit**,** it lifts the valve V with its weight & the excess steam escape through the pipe to the outside. This valve is used only with stationary type of boilers. It is the most elementary type of safety valve. The objection to dead weight safety valve is the heavy weight that has to be carried.

(2) Lever safety valve

The disadvantages of using a heavy weight in a dead weight safety valve are eliminated in a lever safety valve by the use of a lever. The valve rests over the gunmetal seat, which is secured to a block fixed upon the boiler. One end of the lever is hinged while the other end carries a weight. The thrust of the lever with its weight is transmitted to the valve by a short strut. When the steam pressure exceeds the normal limit, it lifts the valve & the lever with its weight. The excess steam will thus come out of the boiler. It is suitable for stationary boilers.

(3) Spring loaded safety valve

There are various types of spring-loaded safety valves in use of which a locomotive safety valve is described here. This is also called a Rams bottom safety valve. It consists of a cast iron body with 2 branch pipes. Two separate valves are placed over the valve sittings, which are secured to the top of the branch pipes. A lever is placed over the valves by means of two pivots. The lever is held tight at its proper position by means of a spring, one end of which is connected with the lever while the other end with the body of the valve. When the steam pressure exceeds the normal limit, the valve rises up against the action of the spring & the steam escapes out. For testing, the valve may be opened by pulling the lever in the downward direction. Spring-loaded safety valves are now widely used for stationary & non-stationary boilers.

(4) High steam and low water safety valve

The valve is generally used at the Cornish or Lancashire boiler. It allows the steam to escape when the steam pressure exceeds the working pressure or when the water level falls below the normal level. So this is a combination of 2 valves used for 2 purposes. The valve V rests on a valve seat & the valve U loaded with the weight W_2 rests on the valve V. When the steam pressure exceeds the normal limit, the valve along with the valve U raises up against the thrust of the lever, & excess steam escapes out. Inside the boiler, a lever L is hinged at a point F. One end of the lever carries a weight W_1 while the other end has a large earthenware float E. The lever is balanced when the weight E is under the water. But when the water level falls below the normal limit, it is unbalanced & its knife-edge K pushes the collar C that is fixed to the valve U having a weight W_2. The valve U raises with the weight W_2 & the steam escapes through a specially constructed passage causing a loud noise.

B. BOILER ACCESSORIES

Accessories are not compulsory for the functioning of a boiler but it can be a part of the boiler layout itself. The function of boiler accessories is to increase the efficiency of the boiler.

The following boiler accessories are generally used.

- Economizer
- Super heater
- Air preheater
- Steam separator
- Steam trap
- Boiler Feed Pump
- Injector

Economizer

A device, which utilizes the waste heat, carried by the exhaust gases before leaving the boiler through chimney or exhaust steam or utilized steam for heating feed water is called economizer. It is generally used for stationary boilers.

It consists of a large no. of vertical cast iron pipes which are connected with 2 horizontal pipes, one at the top & the other at the bottom. The feed water is pumped into the economizer through the bottom pipe. From the bottom pipe the water comes into the top pipe through the vertical pipes & finally comes into the boiler. The water is allowed to pass in a direction opposite to that of the furnace gas.

At the rear end of the vertical pipes, a blow-off valve is fitted through which any mud or sediment deposited in the bottom boxes may be discharged. The soot of the flue gases deposits on the outer surface of the vertical pipes & this reduces the efficiency of the boiler. A system of scrapers is, therefore, adopted to clean the outer surface of vertical pipes. Each vertical pipe is fitted with a scraper which moves up & down automatically & continuously by means of a gear & belt arrangement. A safety valve is also provided at the end of the pipe for the safety of the pipes.

Super heater

The function of a super heater is to increase the temperature of steam with the supply of additional heat. They are located in the path of the furnace gases so that heat is recovered by the super heater from the exhaust gases.

It consists of two mild steel boxes or headers. Group of solid drawn steel tubes bent to U form is hanged from the headers. The ends of these tubes are expanded into the headers. C & D are external & internal covers. For inspection & cleaning, the space is provided between the headers. The covers close this space. Any tube may be taken out & replaced without disturbing the super heater. The branches on the headers for the inlet & outlet of the steam are forged on the headers. The steel flanges have been screwed on to the header. A balance damper is provided in the super heater to prevent the overheating of the super heater tubes. A handle operates this damper. By turning the damper upwards into the vertical position the gases pass directly into the bottom flue without passing over the super heater tubes. By

placing the damper in intermediate position, part of the gases will go over the super heater tubes & the remaining will pass directly to the bottom flue. Varying degrees of superheat may be thus, possible to obtain.

Air preheater

The air preheater heats the air before it enters into the combustion chamber by using the heat of flue gases, which are coming out from the economizer. Now supplying preheated air to the boiler saves some amount of fuel. As we need some amount of heat in removing the moisture from the fuel, the preheated air will supply this heat. Thus that much amount of the heat we have to supply less in to the boiler in form of fuel.

Steam Separator

The steam separators separate the steam and water before the steam enters into the steam turbine. There is a typical requirement of the steam turbine that the steam, which is entering it should be dry. There should not be any water particles in it.

Steam trap

The steam traps are used in the steam power plant in order to remove the water particles from the steam.

Feed pump

The feed pump increases the pressure of feed water, which is coming out from the condenser.

As we know the pressure inside the boiler is higher than the atmospheric pressure and the condenser pressure, so the water will not flow from condenser or from atmosphere automatically. So feed pump is used.

Injector

The injector injects feed water into the boiler. Injectors can be used alternatively in place of feed pumps, as the function of both is same, which is to increase the pressure of feed water to give it entry into the boiler.

QUESTIONS:

1. What is boiler mounting? Give examples of boiler mountings.
2. Write the location & functions of following mountings.
    1. Steam stop valve.
    2. Blow of cock.
    3. Feed check valve.
3. Write the location & functions of following mountings & explain their working with a neat sketch.
    1. Water level indicator.
    2. Fusible plug.
    3. Bourdon tube pressure gauge.
    4. Dead weight safety valve.
4. Why are accessories used in boiler? Give examples of boiler accessories.
5. Write the location & function of following accessories & explain their working with neat diagrams.
    1. Economizer.
    2. Super heater.
    3. Air preheater.
6. Why are scrappers used in economizer?

<u>Conclusion:</u>

Marks obtained:	**Signature of faculty:**	**Date:**

Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO. 3

STUDY OF VAPOUR COMPRESSION REFRIGERATION CYCLE AND SIMPLE WINDOW AIR CONDITIONING SYSTEM

AIM:
To study the Vapour Compression Refrigeration cycle and Simple Window Air conditioning system.

OBJECTIVE:
To know the basics about Refrigeration and air conditioning system.

THEORY:

A. VAPOUR COMPRESSION REFRIGERATION CYCLE

The principal parts of the system and their functions are as under:

Evaporator: To provide a heat transfer surface through which heat can pass from the Refrigerated space or product into the vaporizing refrigerant.

Suction line: To convey the low-pressure vapour from the evaporator to the suction inlet of the compressor.

Compressor: To remove the vapour from the evaporator and to raise the temp. and pr. of the vapour to appoint such that the vapour can be condensed with normally available condensing media.

Discharge line or Hot gas line: To deliver high pressure, high temperature vapour from the discharge valve of the compressor to the condenser.

Condenser: To provide a heat transfer surface through which heat passes from the refrigerant to the atmospheric air.

Receiver: To provide storage for the condensed liquid so that a constant supply of liquid is available to the evaporator as needed.

Liquid line: To carry the liquid refrigerant from the receiver to the refrigerator flow control.

Refrigerant flow control: To meter proper amount of refrigerant to the evaporator and to reduce the pre. of liquid entering the evaporator so that liquid will vaporize in the evaporator at the desired low temp.

B. AIR CONDITIONING SYSTEM

Air Conditioner Basics

An air conditioner is a mechanical device that transfers heat from inside your home to the outside air. Heat is removed by first passing indoor air through an evaporator coil where refrigeration lines containing R-22 absorb the heat. The heat is then carried outside to the condenser coil where it is released into the outside air. This process continues until the air inside your home is at the desired temperature. An air conditioner doesn't just cool the air. There are other functions that can have an equally important role in your comfort.

The four functions of an air conditioner are:

- Cools the air - removes the heat.
- Dehumidifies the air - removes the moisture.
- Filters the air - removes the dust.
- Circulates the air - evenly distributes the air.

If any one of these four functions is not operating properly, you will not feel comfortable. To accomplish the four functions above, every air conditioner must have these basic components:

- Compressor - compresses and pumps the refrigerant through the system.
- Evaporator coil - absorbs heat and moisture from your home.
- Condenser coil - releases the heat into the outside air.
- Fan motor - circulates the air inside your home and also blows hot air outside.
- Refrigerant expansion device - cools the refrigerant so that it can absorb heat.
- Operating controls - turns unit on/off, sets desired temperature, controls fan speed, etc.

Air Conditioning Systems

The system by which air conditioners provide cooling is called the Refrigerant Cycle. This system has four major components common to all air conditioning systems (see figure). These components and their basic functions are listed below.

1. *Compressor* - Refrigerant is drawn from the evaporator and pumped to the condenser by the compressor. The compressor also pressurizes the refrigerant vapor so that it will change state (condense) readily.
2. *Condenser* - The high-pressure refrigerant vapor releases heat through the condenser coils as it condenses into liquid refrigerant.
3. *Metering Device* (capillary tube, expansion valve) - The metering device restricts the flow of liquid refrigerant from the condenser to the evaporator. As refrigerant passes through the metering device, its pressure decreases making it easier to vaporize.
4. *Evaporator* - The low-pressure liquid refrigerant absorbs heat as it vaporizes in the evaporator coils.

Split system (Central Air Conditioning)

The components of this system are separated in two different locations. The evaporator is located inside and provides cooling through a central air duct. The compressor and condenser are located outside. This system is most commonly used for residential applications, small office buildings and commercial buildings.

Window Air Conditioner

This is a small packaged unit designed to fit in a window, primarily residential use. Room air or ambient air is drawn in through the evaporator intake in front. The air is cooled as it passes over the evaporator coils and leaves through the cool air nozzle(s). The condenser intake on the side of the unit draws in air to cool the condenser coils. This warm air is then discharged out the condenser exhaust on top.

QUESTIONS:

1. Define following terms with respect to refrigeration.
   - Refrigeration
   - Ton of Refrigeration
   - Refrigerant
   - Refrigerating effect
   - COP
2. Explain the working of followings with neat sketches. (Name the major components.)
   - Simple Vapour compression refrigeration cycle.
   - Vapour absorption refrigeration system.
3. Why the evaporator is provided at the top portion of the refrigerator?
4. Why the glass is provided on the vegetable compartment?
5. Define air conditioning.
6. Explain the working of Window air conditioning system with neat sketch. (Name the major components.)
7. What is split A/C unit? Why it is required?

Conclusion:

Marks obtained:	Signature of faculty:	Date:

Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO. 4
STUDY OF POWER AND MOTION TRANSMISSION SYSTEMS

AIM:
To study the power and motion transmission systems.

OBJECTIVES:

1. To under stand the different types of power transmission elements, their features and their suitability with respect to given constraint.
2. To known the different types of power transmission drives, their selecting criteria as well as merits and demerits.

THEORY:
The power is transmitted by different modes. One of these modes is mechanical drive. It comprises different elements depending on the requirements which are as under:

1) Shaft or Shafting.
2) Belt.
3) Pulley.
4) Gear.
5) Chain.
6) Bearings and bearing blocks.

<u>Belt</u>
A belt is a flexible element running over pulleys. Which are used mostly in belt drives. They are made with different cross section and from different materials depending on amount of torque which is going to be transmit, the center distance between the driving and driven shaft.

Types of Belt

a) Flat belt

They are rectangular in cross section with standard width and thickness. They are available in the form of long strap which can be join by suitable type of joint with require length. Mostly used when distance between two shafts as large as 10 meters. Through this type of belts we can get the clutch action by shifting it from loose to tight pulley.
The materials for flat belt are leather, rubber, canvas etc.

b) V-belt

They are trapezoidal in cross section with standard dimensions and available in endless form in standard length. They are run in v-grooves made pulleys. The number of belts used is depends on the amount of torque, which is to be transmitting. By using this types of belts slip can be reduce. They are mostly used to connect shafts, which have center distance less than or equal to 5 meters. By this types of belts speed ratio can be achieve up to 7.
The material for v-belts is fabric-vulcanized rubber with cotton or nylon cord tension element.

c) Timing belt

These belts are basically flat belt with teeth molded on inner side. They are available in standard endless length, pitch and width. They are also known as positive drive belts. Mostly used for short center distance arid for higher power transmission. They are comparatively costlier.

The material used for these types of belts is high quality rubber reinforced with the high tensile fabric cords.

d) Rope

It is also one kind of circular belt which is used mostly in factories and workshops, when the distance between shafts is running from 8mtrs. to 30 meters.

The materials used for these are leather, cotton, fabric and rubber and metal Wires. The fiber ropes are used when the pulleys are about 60 meters. Apart, while wire ropes are used when the pulleys are up to 150 meters apart.

Pulleys

The function of this element is to transmit the power from one shaft to the belt which running over it, by giving motion in virtue of friction. They are generally made from C.I, Cast steel, wrought iron and aluminum alloys, wood, nylon, plastics, fibers etc.

Constructional Details:

The main parts of pulleys are as under:

a) Rim: Is an outer most part of pulley. It may flat or grooved or convex on which belt is runs. The convex feature preventing the slipping of belt and it keep in tension to the belt.

b) Hub or Boss: A Central hollow cylindrical portion of pulley called hub, in which shaft is inserted and to prevent relative motion between them by different methods.

c) Arm: It joining the hub and the rim. They may be straight or curved have various cross section like circular or elliptical etc.

d) Rib: It is a circular disc like portion of pulley joining the hub and the rim. Smaller diameter's pulley provided with it.

Types of Pulley

a) Cone pulley (Stepped pulley)

Mostly made from C.I., having several steps of different diameters, it used to get varying velocity ration of power transmission between parallel shafts by simple shifting the belt from one step to other. The driving and driven pulley are placed opposite manner and diameter of the corresponding steps are such that same belt can be used.

b) Guide pulley

Used to guide the belts connecting nonparallel intersecting or nonintersecting shafts. It is also keeps the belt in proper plane.

c) Jockey (Rider) pulley

It increases the arc of contact and also keeps the belt in tension. It mounted near the smaller of the two-transmission pulleys in a belt drive and always positioned to ride on the slack side of the belt.

d) Grooved pulley

It has a groove at rim and cross section of it depends on the cross section of belt or rope. Generally made from C.I. Because of groove, comparative friction is higher so can be used for transmitting higher torque.

e) Fast and loose pulley

Pair of pulley mounted on the shaft out of which one is provided with brass or gunmetal bush and it rotates freely where as other is keyed to shaft. The arrangement enable a machine shaft to be started or stopped at will without stopping the running of the belt by shifting the belt from loose pulley to fast pulley or fast pulley to loose pulley by a suitable belt shifter.

Chain

It consists of several rigid links hinged together to provide on endless flexible loop passing over the driving and driven wheel called sprockets. Sprockets have projecting teeth, which fit into recesses in the link of chain enabling a positive drive.

The pitch can be given by following formula,

$$P = d \cdot \sin\left(\frac{180^\circ}{Z}\right)$$

where, d = Diameter
Z = Number of teeth on sprocket

Roller chain commonly used for power transmission. In bicycles, motor cycles, textile machineries and agricultural equipment the roller chains are frequently used. In context of stands they can be classified like simplex, duplex or triplex chain.

Gear

Gear is a toothed member (link), which is used to transmitting the motion by means of successively engaging teeth from a rotating shaft to another, which rotates, or from a rotating shaft to a body which translates. They are used to transmit power between shafts that are parallel, intersecting or neither parallel nor intersecting by the use of various types of tooth gears.

Types of Gears

a) Spur gear

Used where the power is transmitted between parallel shafts. The teeth may be internal or external. These gears have very wide applications like in watches, measuring instruments, machine tools, automobiles etc. In these types teeth are parallel to shaft axis.

b) Helical gear

These are similar to spur gear except that their teeth are not straight to the shaft axis but follow a helical path, with either left hand helix or right hand helix. They are used to connect parallel shaft as well as nonparallel shaft, nonintersecting shaft. Properly designed helical gears run more smoothly and quietly at high speeds. In this types of gears teeth are gradually engaged with each other because of this, the noise pollution and impact load is low compare to spur gears. The draw back of these gears is axial thrust experience during motion. The solution of this problem can be solved by double helical (Herringbone) gear.

c) Spiral gear

Skewed gears are used to connect two nonparallel, nonintersecting shafts. Their teeth are like helical gear cut along helical path on the wheel periphery. The teeth have point contact. Suitable for small power transmission.

d) Bevel gear

They have their blanks conical in shape and having varying in cross section along the tooth width. They are mostly used where two axis of shaft intersecting in a plane, they may be at right angle or may not be. The bevel gears having angle between shafts at 90° are called miter gears and other then 90° is called bevel gears. Depending on their shape of teeth they may be classified as straight tooth type, spiral or skew or hypoid bevel gears. Spiral has smoother running action. Hypoid bevel gears are used in automobile differential gear box.

e) Worm gear

These are special forms of helical gear in which teeth have line contact and axes of driving and driven shaft are at right angle and nonintersecting. A smaller gear is called worm and large one called worm wheel. This combination use for very large velocity ratio and for law space. The noise and vibration are cooperative low. Their common application are in lath, drilling, milling etc. They are also used in reduction gearbox.

f) Rake and pinion

This may consider as a pair of spur gear having one of the gear having infinite pitch circle diameter (PCD) is called rack. Where as other small gear is known as pinion. These are used to transmit circular motion into rectilinear motion and vice- versa. They are fitted in lathe, drill, planner etc. to convert rotary motion into transitory motion.

<u>Power transmission drives</u>

As we know power is transmitted by different type of drives which are framed in following tree diagram.

MECHANICAL DRIVE

- Frictional drives
  - Belt drive
  - Rope drive
- Positive drives / Mesh drives
  - Chain drive
  - Gear drive

Frictional drive

In this type the power is transmitted by the virtue of friction. If the distance between source and utilization place is small than direct drive is used other wise flexible drive is used.

BELT AND ROPE DRIVE

To transmit the power from one shaft to another shaft, pulleys are mounted on two shafts. Which are connected by belt or a rope depends upon the requirement. The belt /rope should be in tension so that motion of one pulley is transmitted to other. Through this system constant velocity ratio can not be maintained because of slipping and straining of belt/rope. Diameters of driving and driven pulley are the controlling parameter for the velocity ratio and their relation can be written as under,

$$\frac{N_2}{N_1} = \frac{D_1}{D_2}$$

Where, N_1, N_2 = speed of pulleys.
D_1 , D_2 = respective diameter of pulleys.

With respect to belt position and relation between axes of the shaft, belt drive can be classified.

a) Open belt drive

When shaft axes are parallel and rotate in same direction. In this case driver pulls the belt from one side and delivers it to other side. Thus the tension in the lower side will be more than that in upper side. The lower side known as tight side where as upper side called slack side.

b) Cross belt drive

Used where the shaft axes are parallel but direction of rotation of two shafts is in opposite direction. In this case driver pulls the belt from one side and delivers to other side.

c) Quarter turn drive

Used for nonparallel non-intersecting shafts. Provided that the center line of the belt when approaches a pulleys lies in the mid plane of that pulley.

d) Right angle drive

Used when axes of shaft are at right angle. In this one additional suitable guide pulley used. This is directing and can be connect to belt.

POSITIVE DRIVE

To avoid slipping and for getting constant velocity ratio this type is used. In this projecting portions (tooth) are mash with corresponding recesses. This constrained to move together without slipping and ensure perfect velocity ratio (V.R). Under this class chain drive and gear drives are falling which are discuss as under:

<u>Gear trains</u>

An assembly of gear wheels by means of which motion is transmitted from one shaft to another shaft is called gear train. It may include any or all kinds of gears, depends on the relative axial relationship between the shafts and the require V.R .The ratio of rpm of driver (1^{st}) gear to rpm of the driven (last) gear is known as speed ratio or V.R of gear train. It will be positive if direction of rotation 1^{st} and last is same and negative in case of opposite. As we have seen that depending on the gear arrangement different type of gear train can be framed as under,

a) Simple gear train

A series of gears, capable of receiving and transmitting motion from one gear to another is called simple gear train. In it, all the gear axes remain fixed relative to the frame and each gear is on separate shaft. In this type,

1) A pair of mating external gear always moves in opposite direction and move in same direction in case of one internal gear (annular) and one external pair.
2) All odd number gears are move in one direction and all even number gear move in the opposite direction.
3) Shaft axes are parallel to each other.
4) Intermediate gear does not effect on V.R and therefore they called as idlers. Some time idler may use to change the direction of driven gear. Their formulas are as under,

$$\text{Train value} = \frac{\text{Rpm of driven}}{\text{Rpm of driving}} = \frac{\text{Number of teeth of driving}}{\text{Number of teeth of driven}}$$

$$\text{V.R.} = \frac{1}{\text{Train value}}$$

b) Compound gear train

When two or more than two gear are mounted on common shaft it is known as compound gear. It gives compact layout. The train value of type can be denoted as under,

$$\text{Train value} = \frac{\text{Product of number of teeth of driving gears}}{\text{Product of number of teeth of driven gears}}$$

c) Reverted gear train

First and last gears run at same axis, it is also one kind of compound train. Very large reduction ratio can be achieved in compact place by this type. The speed ratio and their pitch circle radius relation can be written as under,

$$\text{Train value} = \frac{\text{Product of number of teeth of driving gears}}{\text{Product of number of teeth of driven gears}}$$

$$R_1 + R_2 = R_3 + R_4$$

Where, R_1 = radius of driving gear
R_2, R_3 = Intermediate gears
R_4 = radius of driven gear

d) Epicyclic gear train

When there exists a relative motion of axis in gear train, it is called a planetary or epicyclic gear train, thus in an epicyclic train the axis of at least one of the gear also moves relative to the frame. If the arm is fixed, the gear train becomes a simple gear train. Large speed reduction is possible with this type and if the fixed wheel is annular a more compact unit could be obtained. Important applications of epicyclic gear train are in transmission, computing devices etc.

QUESTIONS:

1. Explain methods of drive and differentiate it.
2. Explain with neat sketch the working of various types of belt drives used for motion and transmission. Also give its merits and demerits.
3. State different types of pulleys used in power transmission and explain fast and loose pulleys with neat sketches.
4. Explain various types of gear with neat sketches.
5. Define gear train and give the classification of gear train with neat sketch.

Conclusion:

Marks obtained:	Signature of faculty:	Date:

Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO. 5
STUDY OF COUPLINGS, CLUTCHES AND BRAKES

AIM:
To study the different types of couplings, clutches & brakes.

OBJECTIVES:
1. To know the working principle of couplings, clutches and brakes.
2. To know the constructional features and their functional importance in particular items.
3. Their applications in industrial field.

THEORY:

In practice it is rare that power generation and power utilization at the same place. It is transformed in different mode. Here we are limiting our discussion up to mechanical mode. Transmission nature is a preferential and with respect to this different elements are used.

When mechanical power transmitted by continues rotating motion and input & out put members require acting as same unit in that situation members are join t couplings. Where as requirement is discontinue / continue and with limiting extremity (power flow. In that constrains mostly clutches are used and for control over power value on input or output member brakes are used, They are used to slow down, stop or hold a load or release a load and control its speed.

Couplings

Most common power transmitting member is shaft, which limited by manufacturing and shipping requirements. The elements which joint two shifts are know as couplings, which transmit power one to another. On the bases of axial relationship between two shafts, different types of couplings are developed which are as under:

1. SLEEVE COUPLING

It consist of sleeve or hollow cylinder normally made from cast iron, which receives end of two shafts made to butt together inside the sleeve. Sleeve and shaft are keyed by sunk key, which normally made from same shaft material. Some time two keys may be a used. Torque is transmitted from driver shaft to sleeve and from sleeve to driven shaft. It has no projected parts and simple constructional features still require careful fitting.

2. SPLIT MUFF COUPLING

It is an improved form of sleeve coupling. In this sleeve is split into two halves and connected by bolts. Here key also used as in sleeve coupling to create a pressure between the surface of shaft and the sleeve there is some clearance at joining of to sleeve halves. So tightening of the bolts causes frictional force between the surfaces. Because of this it can be used for heavy-duty work.

3. FLANGE COUPLING

It is most widely used from medium to heavy duty. It consists of two coupling halves keyed to two shafts, which are coupled by bolts. When bolt heads and nuts protrude beyond the flat surfaces of flanges, the coupling called as unprotected type, which is dangerous. To over come this flange is provided with a shroud or an annular projection, which shelter the bolt heads and nuts. To ensure true alignment, one shaft may entered into other flange or projected portion of one flange fit into the corresponding recess of other flange.

4. BUSH PIN TYPE FLANGE COUPLING

It is a modified form of protected type flange coupling. There is a cushioning effect due to leather/rubber bush in one of the flange. In which enlarged diameter of pins are inserted and other end of bolt is fastened by nut in other flange. This feature is offering a small axial movement and takes care of shock and slight misalignment. It is mostly used for direct connection of electric motor.

5. OLDHAM COUPLING

It is used in connecting two parallel shafts whose axes are at small offset. It consist of three members, two flanges and one intermediate disc, both flange have diametrical slot at right angle to each other, where as intermediate disc has replica of that slots on respective either sides. The rotation of drive shafts causes the rotation and sliding of circular disc, which transmit the power to the driven shaft.

6. UNIVERSAL COUPLING

It is used where the axes of shafts are intersecting to each other and during operation; angle between the axes may slightly very. It consists of forks keyed to the ends of either shaft and they are pin joined to a center block, which has two arms at right angles to each other. This is widely used in automobiles to connect propeller shaft and also in machine tools.

Clutches

Element which used to connect the one rotational part to another stationary, which later is to be set into motion and transmit as well as control the limit of flow of mechanical power, where the power at driven may be required to start or stop frequently. There are four categories: 1) mechanical type 2) electromagnetic type 3) pneumatic type 4) hydraulic type.

We will study here **mechanical type clutches**.

JAW AND TOOTHED TYPE CLUTCH

Positive clutches used to connect driven shaft to prime mover when it is not objectionable to start the driven suddenly and where the connected load (including inertia resistance) is not so great, which is also not so harmful to the prime mover it self. It consists of segmental projection or dogs on one side flange and corresponding recesses on other side flange on the driven shaft. In this driving flange is rigidly attached where as driven flange is keyed by feather key so it can slide on that shaft. Some time involute teeth also provided.

FRICTIONAL CLUTCH

Power transmitted by virtue of friction between the surfaces. These are important when engagement must be made under load and prime mover is to be protected against sudden load. Apart from this, shafts running at different speed are to connect, without stopping. These features make them suitable for vehicles, process machineries, machine tool and where the frequently power cut-off either partial or completely required. Frictional surfaces may be flat, conical or cylindrical and they are named with respect to that.

a) Disc clutch: In this type, one flange is rigidly keyed to driving shaft. Where as other is fitted on driven by splines or feather key so it can move along the axis of shaft. During working movable flange is pressed by mechanism so the torque is transmitted by friction through the disc, which is between two flanges. In this type of clutches the amount of transmitted torque will be depends on axial pressure, contact area of surfaces and coefficient of friction. On the bases of number of disc frictional clutches can be classified like single disc clutches, multi disc clutches. When large torque is to be transmitted at that time multi disc clutches is preferable, because of higher number of contact surfaces.

b) Cone clutch: In this contact surfaces of flange are in conical shape. The slop of cone is varying from 8° to 15°, the clutch parts (cones) are held together by springs, which produce axial force. In this type of clutches the excessive axial pressure is not required.

c) Centrifugal clutch: They are used in a wide variety of application to connect drives (electric motor, I.C.engines etc.) In this type of clutches power flow is possible only after reaching certain amount of speed of driving shaft. The engagement of shoes with the inner lining of drum will take place when centrifugal force becomes greater than spring force. This engagement will cause the transmission of torque from driving to driven.

BRAKES

It is a element whose primary purpose is to absorb a kinetic energy by way of frictional resistance. Which are generally used for slow or stopping the moving parts, there are some brakes which are used to with hold conversion from potential to kinetic energy of the object as they are lowered by hoists elevators etc. Other main function is to hold a body in rest of uniform motion against accelerating forces/ couples and preventing unwanted reversal of the direction of rotation. The major functional different between clutch and brake is that unlike a brake, which connects a moving member to a stationary frame, a clutch connects a rotating member to another rotating member. In brakes absorbed energy dissipate in form of heat.

They are used in automobile, trains, vehicles, hoists elevators, press and in machines.
Their different classes are:
Mechanical type brakes, electrical type brakes, hydraulic type brakes and some other types. Here our discussion is limited up to **mechanical** types.

Types of Brakes

a) Block brake: This brake consists of block, which are pressed against the rim of revolving brake wheel or drum, which faced with frictional material. The blocks are made of soft material compare to drum or wheel, like wood or soft metal.

These types of brakes are mostly used in railway trains, tram cars etc. Prony brake is also belongs to this class.

b) Band brake: It consists of flexible bend of leather of steel lined with friction material, which embraces a part of the circumference of brake drum. In context of interrelated position between band ends and fulcrum, band brake may be simple or differential type band brake. When force is applied at lever end the friction is generated between band and drum surfaces, which cause speed reduction or stopping of wheel.

The band and block type brake is improved type of band brake, which consists of number of wooden blocks fixed to a steel band.

Properties required for brake linings

1) High mechanical strength.

2) High coefficient of friction.

3) Law wear rate.

4) High heat dissipation capacity.

5) Should not affect by moisture or oils.

Examples: wood, leather, asbestos and ferrodo.

c) Internal expanding shoe brake: Widely used in automobiles. This may be operated by mechanically or hydraulically. It consists of two shoes lined with some friction material, pivoted at one ends about a fixed fulcrum. The shoes are housed inside the rotating drum. Under release condition there is a gap between shoes and inner surface of the drum by spring, when shoes are expanded by mechanically /hydraulically. The friction between shoes and the drum produce a braking torque and hence reduces the speed of the drum. These types of brakes are mostly used in automobiles

QUESTIONS:

(1) State the functions of couplings, clutches and brakes.
(2) Classify couplings, clutches and brakes.
(3) Describe in brief a flange coupling with a neat sketch. State why flexible couplings are preferred to rigid couplings.
(4) Distinguish between a clutch and coupling and brake.
(5) Writes short notes on followings with neat diagrams:
1. Universal coupling.
2. Oldham coupling.
3. Centrifugal clutch.
4. Internal expanding shoe brake.
5. Block Brake

Conclusion:

Marks obtained:	Signature of faculty:	Date:

Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO: 6

INTRODUCTION TO SURVEYING AND BASIC SURVEYING INSTRUMENTS

OBJECTIVE:

To understand the working and fundamentals principle of surveying instruments.

THEORETICAL BACKGROUND:

Definition:

Surveying is an art of determining relative positions on, above or beneath the surface of the earth by measuring the horizontal distances & angles for determination of any point on ground.

Object of Surveying:

The object of surveying is to prepare a map / plan, with suitable scale, to show the relative positions of the objects on the surface of the earth.

(i) Map
(ii) Plan

Primary Divisions of Surveying:

(i) Plane Surveying
(ii) Geodetic Surveying

Principles of Surveying:

(i) Working from whole to part not part to whole
(ii) To locate the point by two reference points.

Classification of Surveying:

A. Based on Nature of Field:
    (i) Land
    (ii) Hydrographic
    (iii) Astronomical
    (iv) Aerial
B. Based on Object/ Purpose:
    (i) Geological survey
    (ii) Mine survey

(iii) Archeological survey
(iv) Military survey

C. Based on Instruments:
- (i) Chain survey
- (ii) Compass survey
- (iii) Chain & Compass survey
- (iv) Plane Table survey
- (v) Theodolite Survey
- (vi) Tacheometry Survey
- (vii) Leveling Survey
- (viii) Photogrammetric survey
- (ix) EDM (Electromagnetic Distance Measurement) Survey

D. Based on Methods:
- (i) Triangulation
- (ii) Traversing

WRITE IN BRIEF ABOUT THE FOLLOWING:

Map :
Plan:
Primary Divisions of Surveying:
Plane Surveying:

Geodetic Surveying:

PRINCIPLES OF SURVEYING:

Working from whole to part not part to whole:

To locate the point by two reference points.

CLASSIFICATION OF SURVEYING: Based on Nature of Field:
Land Surveying :
Hydrographic Surveying :
Astronomical Surveying :
Aerial Surveying :
CLASSIFICATION OF SURVEYING: Based on Object/ Purpose:
Geological survey
Mine survey
Archeological survey
Military survey

CLASSIFICATION OF SURVEYING: Based on Instruments:
Chain survey
Compass survey
Chain & Compass survey
Plane Table survey
Theodolite Survey
Tacheometry Survey
Leveling Survey
Photogrammetric survey
EDM (Electromagnetic Distance Measurement) Survey

CLASSIFICATION OF SURVEYING: Based on Methods:
Triangulation
Traversing

INSTRUMENTS USED IN SURVEYING :

(A) For Chaining:

(i) Chains

1. Metric chain
2. Gunter's or Surveyor chain
3. Engineer's chain
4. Revenue chain
5. Steel band

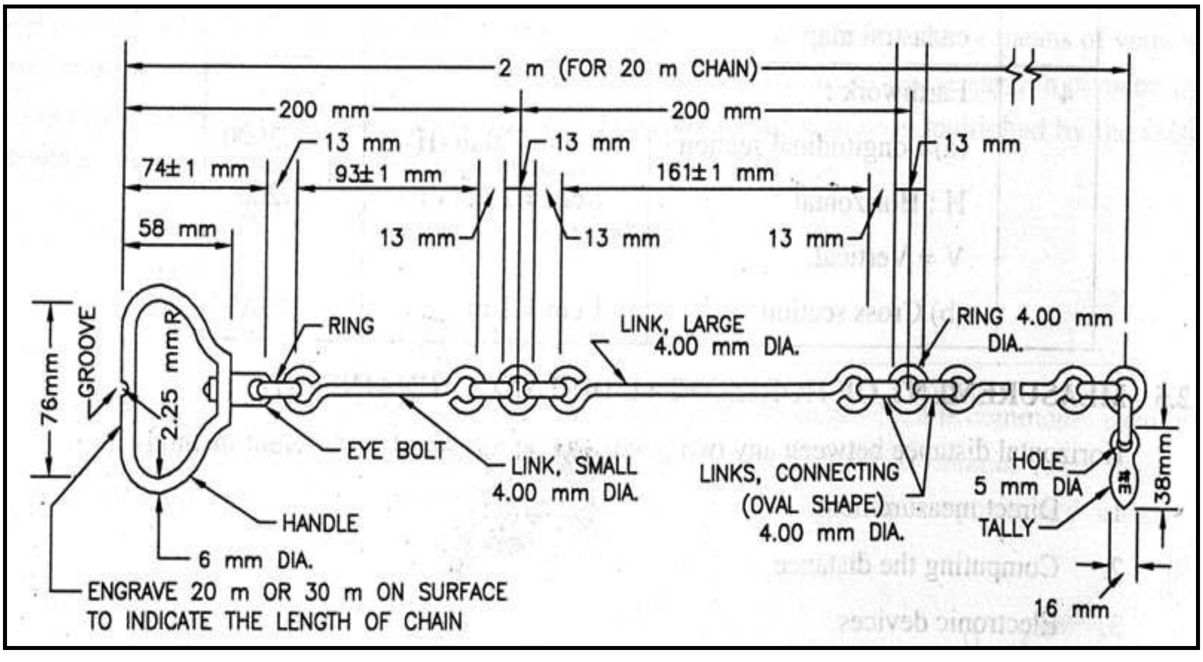


Figure: Details Of Metric Chain

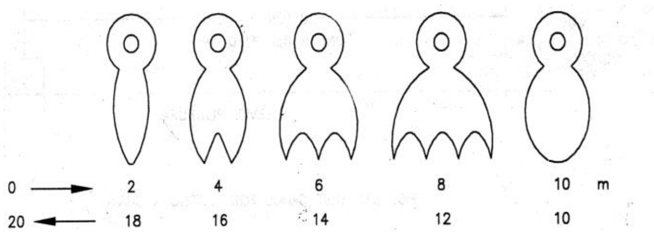


Figure: Tallies for a 20 m chain

Draw the line diagram of chain and tallies
Write in brief about construction of a metric chain
Write in brief about working with a chain

(ii) Tapes

1. Cloth or linen tape
2. Fiber tape
3. Metallic tape
4. Steel tape
5. Invar tape

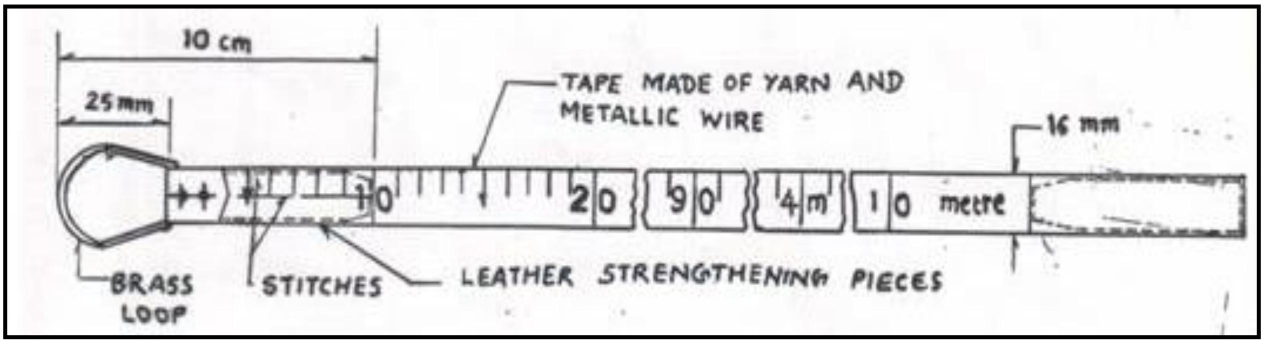


Figure : Measuring tape

Draw the line diagram of Metallic Tape
Write in brief about construction of a Metallic Tape
Write in brief about working with Metallic Tape

(iii)Arrows

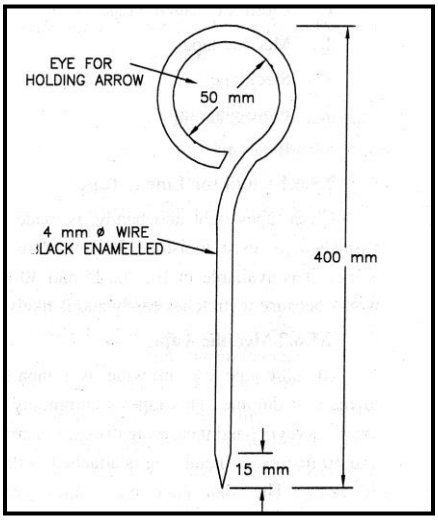


Figure : Steel Arrow

(i) Peg

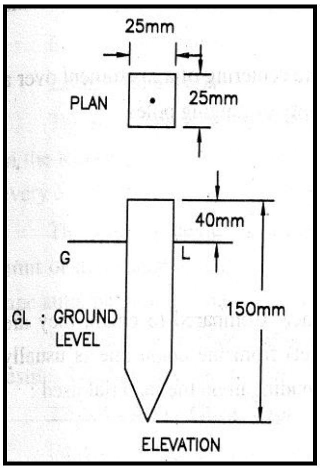


Figure : Wooden Peg

(ii) Plumb bob

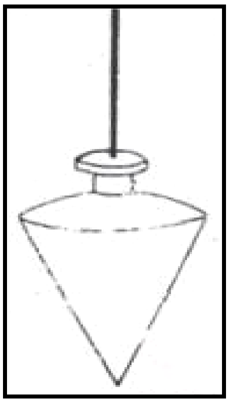


Figure :Plumb bob

Draw the line diagrams		
Steel Arrow	**Wooden Peg**	**Plumb bob**
Write in brief about construction		
Steel Arrow	**Wooden Peg**	**Plumb bob**
Write in brief about working		
Steel Arrow	**Wooden Peg**	**Plumb bob**

(B) For Ranging:

(i) Ranging rod or Ranging Pole

(ii) Offset rod

(iii)Line ranger

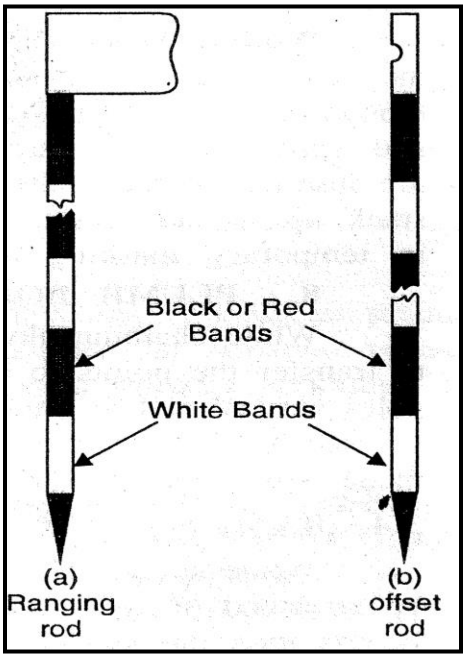
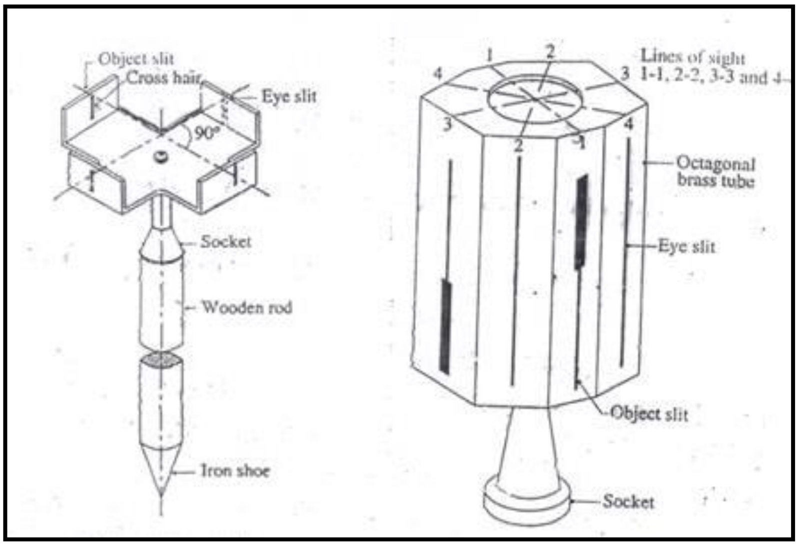


Figure : Ranging rod

Draw the line diagram of Ranging Rod	**Write in brief about construction Ranging Rod**
Write in brief about working Ranging Rod	

(C) For Offsetting:

(i) Cross staff



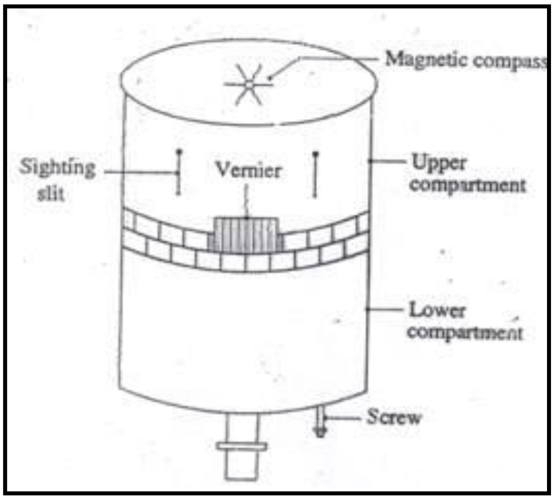


Figure :Cross staff

Draw the line diagram of Cross staff
Write in brief about construction of a Cross staff
Write in brief about working with Cross staff

(iii) Prism square

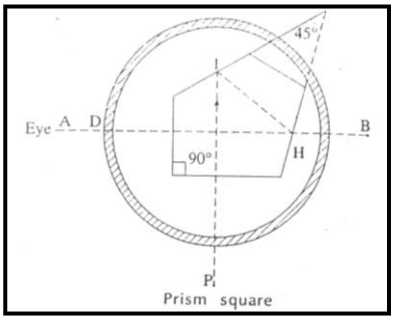


Figure :Prism square

(iv) Optical square

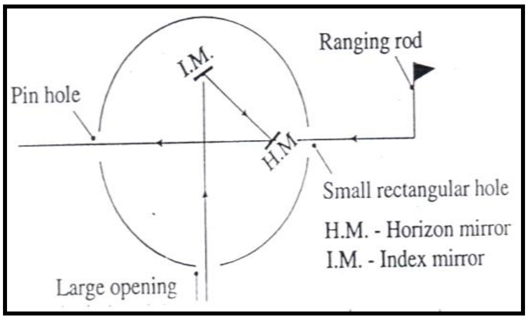


Figure :Optical square

Draw the line diagram	
Prism square	**Optical square**
Write in brief about construction	
Write in brief about working with	

Units of Measure:

Units of Length		
10 millimeters	=	1 centimeter
10 centimeters	=	1 decimeter
10 decimeters	=	1 meter
10 meters	=	1 decameter
10 decameters	=	1 hectometer
10 hectometer	=	1 kilometer
1 foot	=	12 inches
1 inch	=	2.54 cm
1 meter	=	3.28 feet
1 yard	=	3 feet
1 meter	=	1.093 yard
1 mile	=	1.6093 km
Units of Area		
100 sq. millimeters	=	1 sq. centimeter
100 sq. centimeters	=	1 sq. decimeter
100 sq. decimeters	=	1 sq. meter
100 sq. meters	=	1 sq. decameter or 1 are
100 sq. decameters or 1 are	=	1 sq. hectometer or 1 hectare
100 sq. hectometers	=	1 sq. kilometer
1 sq. mile	=	640 acres
1 acre	=	40 gunthas
1 acre	=	10 sq. chain (Gunters)
1 gunthas	=	1089 sq. ft
Units of Volume		
1000 cu. Millimeters	=	1 cu. Centimeter
1000 cu. Centimeters	=	1 cu. Decimeter
1000 cu. Decimeters	=	1 cu. Meter
1 cu. Meter	=	219.969 gallons
1 cu. M	=	1 kilolitre
1 acre. Feet	=	271327 gallons

Marks Obtained:	**Signature of Faculty:**	**Date:**

Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO: 7

LINEAR MEASUREMENT-CHAINING, RANGING & OFFSETTING

OBJECTIVE:
To understand the method of linear measurements through chain survey.
To do Chaining, ranging and offsetting a survey line AB on a level ground.

APPARATUS:
Metric Chain - _____, Measuring Tape - _____, Wooden Pegs, Hammer, Ranging Rods, Steel Arrows.

THEORETICAL BACKGROUND:

Chain Surveying or Chain triangulation:

It is method of surveying in which the area is divided into network of triangles and the sides of the triangle are directly measured on the field with a chain or a tape and no angular measurements are taken.

This type of surveying is used for smaller extent of area with simple details and ground is fairly level.

Terms related with Chain Surveying:

i. Main station
ii. Subsidiary or Tie station
iii. Base line
iv. Survey line
v. Check line
vi. Tie line

Basic Operations in Chain Surveying:

(i) Chaining

The method of taking measurements with the help of a chain or a tape is termed as chaining.

(ii) Ranging

When the length of a line exceeds the length of the chain, to proceed in a straight line, intermediate points are needed to be established between the two stations. The process of establishing intermediate points on a straight line between two endpoints is known as ranging.

Methods of ranging

(i) Direct Ranging

- By Eye
- By Line Ranger

(ii) Indirect Ranging or Reciprocal Ranging

(iii) Offsetting

The method of locating (positioning) objects from the survey line by means of linear measurement (distance between survey line and object), right or left of the survey line is called as offsetting and lateral measurements (distances) are called offsets.

Types of Offsetting:

(a) Perpendicular offset and (b) Oblique offset

PROCEDURE:

1. Main stations A and B are first marked with pegs and ranging rods to make them visible.
2. Unfolding of chain is done and is laid between the station points to measure correct length of the line.
3. The chainman at the station A (At start of a line) is called as follower and who drag the chain forward along the line is called as leader. The follower places zero handle of the chain in contact with the peg at the beginning of the line and firmly holds it. The leader, taking ten arrows and a ranging rod in one hand and other handle of chain in other hand, walks off in forward direction.
4. If the length of the survey line is greater than one chain length, intermediate points are located, in order to pull the chain along a straight line. At the end of chain length, leader holds the rod vertically, approximately in line and then faces the follower. When intermediate ranging rod M is fixed on a straight line by direct observation, the process is known as direct ranging which is explained below.
5. The follower stands half a meter behind ranging rod at A by looking towards the line AB. The follower directs the leader to move ranging rod to the left or right until the

three ranging rods at A, M and B come exactly at same straight line. The code of signals are used as below:

Sr. No.	Signals Given by the Follower	Meaning of the Signal
1	Rapid sweep with right hand	Move considerably to the right
2	Rapid sweep with left hand	Move considerably to the left
3	Slow sweep with right hand	Move slowly to the right
4	Slow sweep with left hand	Move slowly to the left
5	Right arm extended	Continue to move to left
6	Left arm extended	Continue to move to left
7	Right arm up and move to the right	Plumb the rod to the right
8	Right arm up and move to the left	Plumb the rod to the left
9	Both hands above head and brought down	Ranging is correct
10	Both arms extended forward horizontally and the hands brought down quickly.	Fix the ranging rod

6. The leader then marks the position of the rod by making a hole with it. Taking the handle in both hands and standing in line, leader straightens the chain by jerking and stretches it over the mark. Care should be taken that chain should be raised above the ground and should be straight as possible.
7. Leader inserts arrow against end of the chain in the ground, to mark the end of the chain. The leader then picks up other arrows, ranging rod, and drags the chain in forward direction. The leader goes on fixing arrows at the end of each and every chain length. After the tenth arrow is inserted (completing 10 m length), the follower collects all the arrows fixed by the leader and marks the end of the 10^{th} chain by erecting a ranging rod.
8. At the end of the work, as a check the number of arrows inserted is multiplied by chain length which is total length measured.
9. Fractional length of chain is found out by counting the links and multiplying it to size of each link. Fractional length is added to total length to find the true total length.

10. Location of prominent objects can be located by offsetting on a chain line established above. The offsets are usually measured with tape.
11. When offset is to be taken from chain line to the object, say P, the leader holds the zero end of tape at P and follower carrying tape box, swing it along the chain. Wherever the tape shows minimum length is considered to be foot of perpendicular (p), from point P at object, to chain line.
12. Measure Ap and pP.
13. Finally record the survey line AB and offsets in the field book.

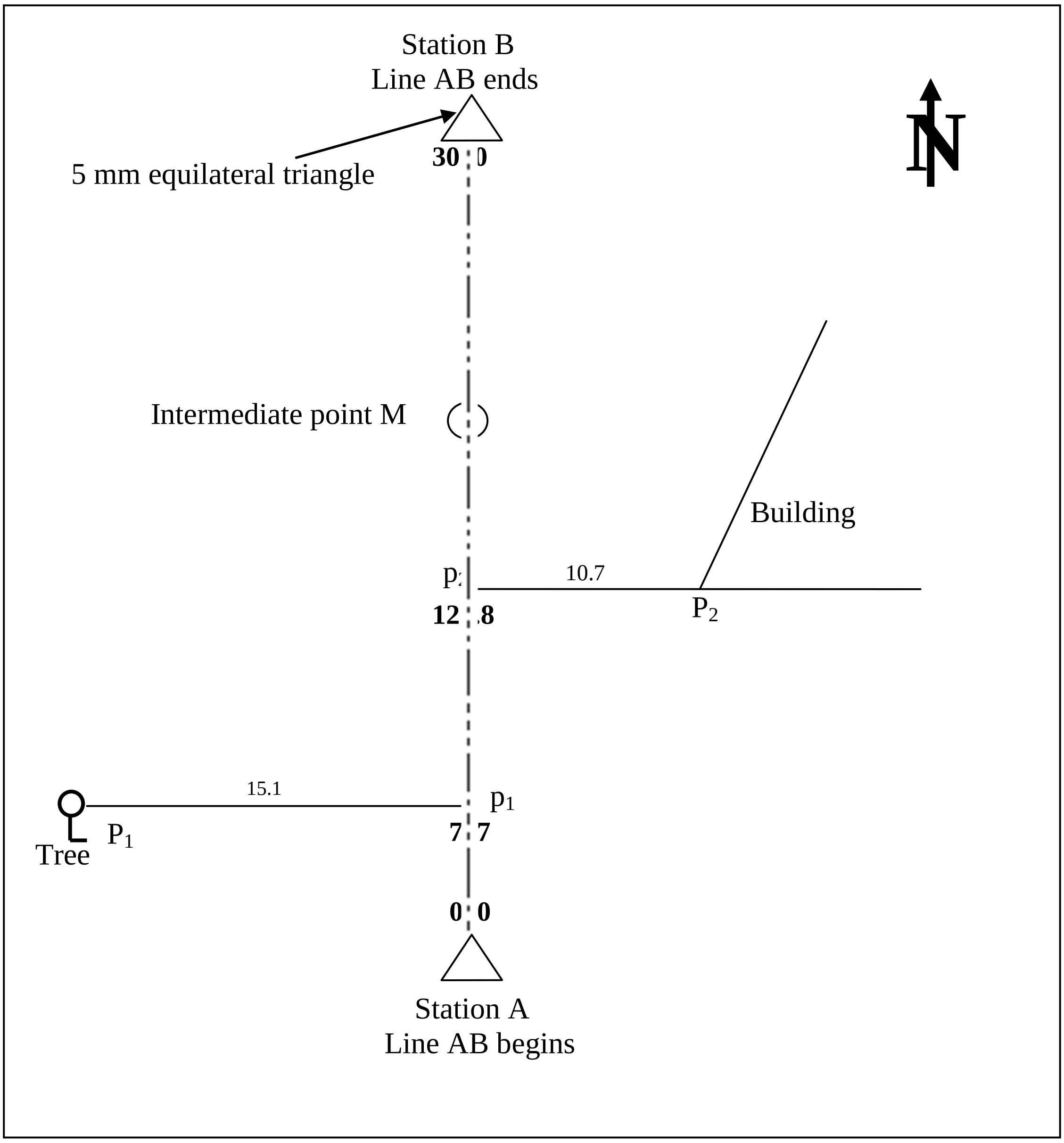


Figure : Representation of a field book entry page

FIELD BOOK ENTRY PAGE

CONVENTIONAL SYMBOLS

Sr. No.	Object	Symbol
1.	North Line	
2.	Main stations	
3.	Traverse stations or sub stations	
4.	Chain line	
5.	**River**	
6.	**Canal**	
7.	**Open Well**	
8.	**Tube Well**	
9.	**Railway Line (single)**	
10.	**Railway line (Double)**	
11.	**Road Bridge or culvert**	
12.	**Railway Bridge or culvert**	
13.	**Road & Rail level Crossing**	
14.	**Wall with gate**	
15.	**Building (Pakka)**	

16.	**Building (Katcha)**	
17.	**Temple**	
18.	**Bench Mark**	
19.	**Tree**	
20.	**Cultivated Road**	
21.	**Embankment**	
22.	**Cutting**	
23.	**Telephone Line**	
24.	**Telegraph Post**	
25.	**Electric Line**	
26.	**Electric Post**	
27.	**Burial Ground or Cemetery**	

Marks Obtained:	**Signature of Faculty:**	**Date:**

Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO: 8

CONSTRUCTION SITE VISIT

OBJECTIVE:

To visit a residential building / educational building and to study about various components of the building

THEORETICAL BACKGROUND:

Types of Building:

(A) Based on Occupancy as per National Building Code:

(i) Residential building

(ii) Educational building

(iii) Institutional building

(iv) Assembly building

(v) Business building

(vi) Mercantile building

(vii) Industrial building

(viii) Storage building

(ix) Hazardous building

(B) Based on Structure

(i) Load bearing structure

(ii) Framed structure

(iii) Composite structure

Components of Building:

Sr. No.	Components of Building	Functions
(A)	**Sub-structure**	
1	Foundation	
(B)	**Super-structure**	
1	Plinth	
2	Damp Proof Course	
3	Walls	
4	Floor	
5	Column	
6	Beam	
7	Opening in walls	
	(a) Doors	
	(b) Windows	
	(c) Ventilators	
8	Sill	
9	Lintel	
10	Weather shed (Chhajja)	

11	Ceiling	
12	Roof	
13	Steps, Stair case & Lift	
14	Finishes for wall	
	(a) Plastering	
	(b) Painting/Varnishing/Polishing	
	(c) White washing	
15	Utility Fixtures	

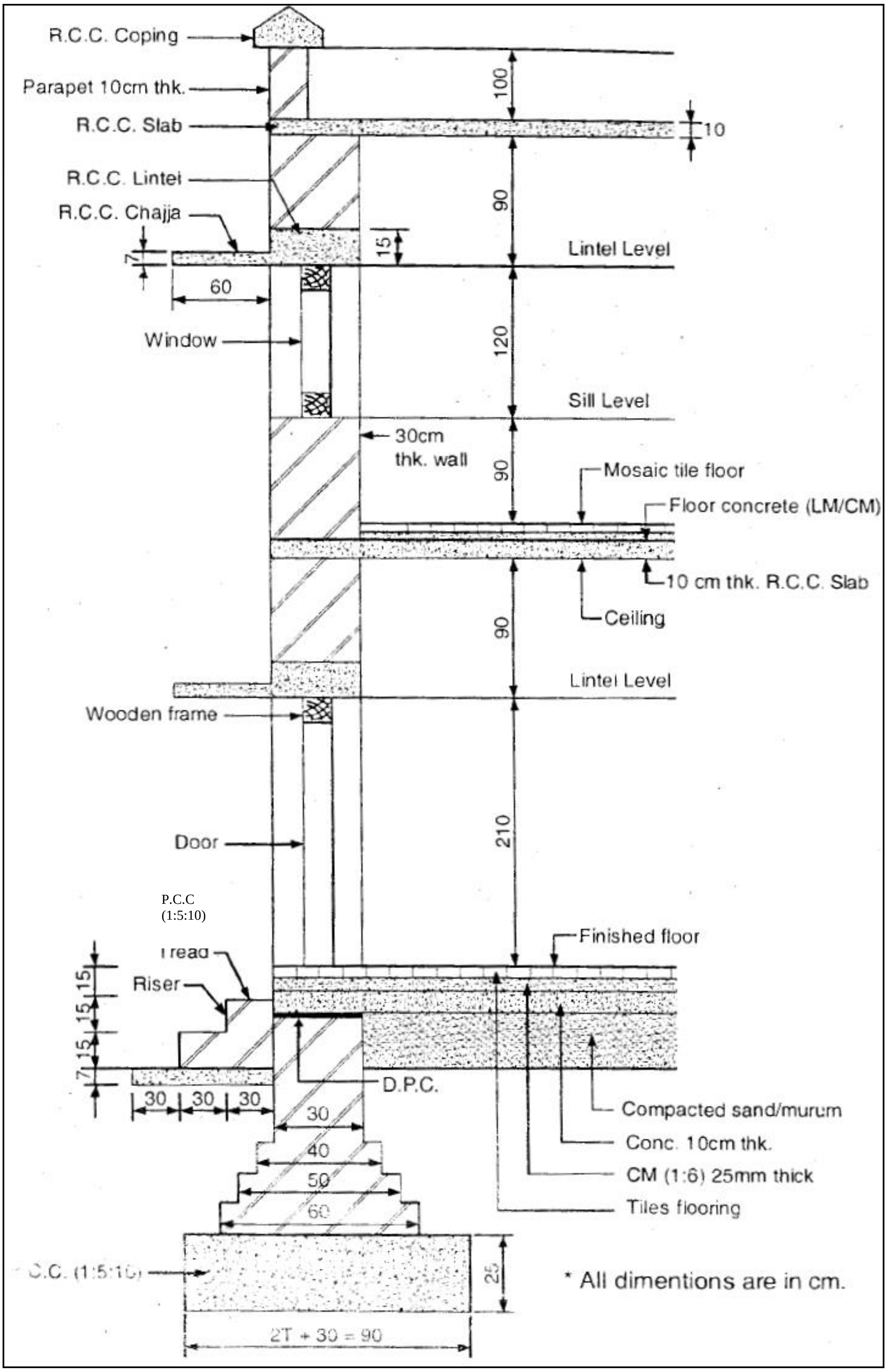


Figure: BUILDING COMPONENTS

(SECTION OF A 30 CM THICK MASONARY WALL)

Types of Building:	
(A)	**Based on Occupancy as per National Building Code:**
(i)	Residential building
(ii)	Educational building
(iii)	Institutional building
(iv)	Assembly building
(v)	Business building

(vi)	Mercantile building
(vii)	Industrial building
(viii)	Storage building
(ix)	Hazardous building
(B)	**Based on Structure**
(i)	Load bearing structure

(ii)	Framed structure
(iii)	Composite structure

VISIT TO A CONSTRUCTION SITE :

- **Observing and learning about ongoing construction activities :**
- **Seeing Different building components :**
- **Collecting and learning about different building materials :**
- **Observation on different construction tools and their usage :**
- **On field group discussion on the building construction :**

A REPORT ON VISIT TO A CONSTRUCTION SITE :

Date of visit :

Name of the place visited :

Ongoing construction activities :

Building components seen on site :

Building materials seen on site :

Construction tools seen on site :

Remarks :

Marks Obtained:	Signature of Faculty:	Date:

Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO: 9

PLANNING AND DRAWING OF RESIDENTIAL BUILDING

OBJECTIVE:

To study principles of planning and to draw the layout of residential building.

THEORETICAL BACKGROUND:

Plan:

Plan is a sort of top view drawn at a predetermined level. It is generally prepared at sill level to get all details of doors and windows. To draw a plan, a building is imagined to be cut in horizontal plane at sill level and then it is seen from top with all the details.

Elevation:

Elevation of building is drawn in vertical plane starting from ground level to roof level. Elevation may be front elevation or side elevation. Elevations are generally drawn above the plan by projecting vertical lines from plan starting from left end side to right end side of the building.

Section / Sectional Elevation:

Sectional elevation is the elevation of the building at a particular assumed section.

Basic Requirements of a Building:

(i) Strength and stability
(ii) Comfort and convenience
(iii) Damp prevention
(iv) Fire protection
(v) Thermal insulation
(vi) Day lighting and ventilation
(v) Sound proofness
(vi) Durability
(vii) Security against burglary
(viii) Termite proofing
(x) Building economics

Principle of Planning:

(i) Aspect

(ii) Prospect

(iii) Privacy

(iv) Furniture requirement

(v) Roominess

(vi) Grouping

(v) Circulation

(vi) Sanitation

(vii) Flexibility

(viii) Elegance

(ix) Economy

(x) Practical consideration

PROCEDURE OF DRAWING:

1. Take appropriate scale to draw the plan, elevation and section for given residential building.
2. The dimensions of line diagram are inside to inside dimensions.
3. Dimensions of the rooms are given as 'Breadth x Length'.
   Breadth in x-direction, Length in y-direction
4. Wall thickness is to be considered 30 cm for main load bearing walls and 15 cm / 20 cm for partition walls.
5. Openings of the building, i.e., doors, windows and ventilators are to be provided as per their location. The sizes of all the openings are not given, they should be assumed of standard size.
   Door (breath x height), Window (breath x height)
6. To draw the elevation and section, chhajja / weathersheds are to be provided over openings located on outer walls of the building.
7. When the heights or levels are not specified, then the standard dimensions for various building units should be assumed.

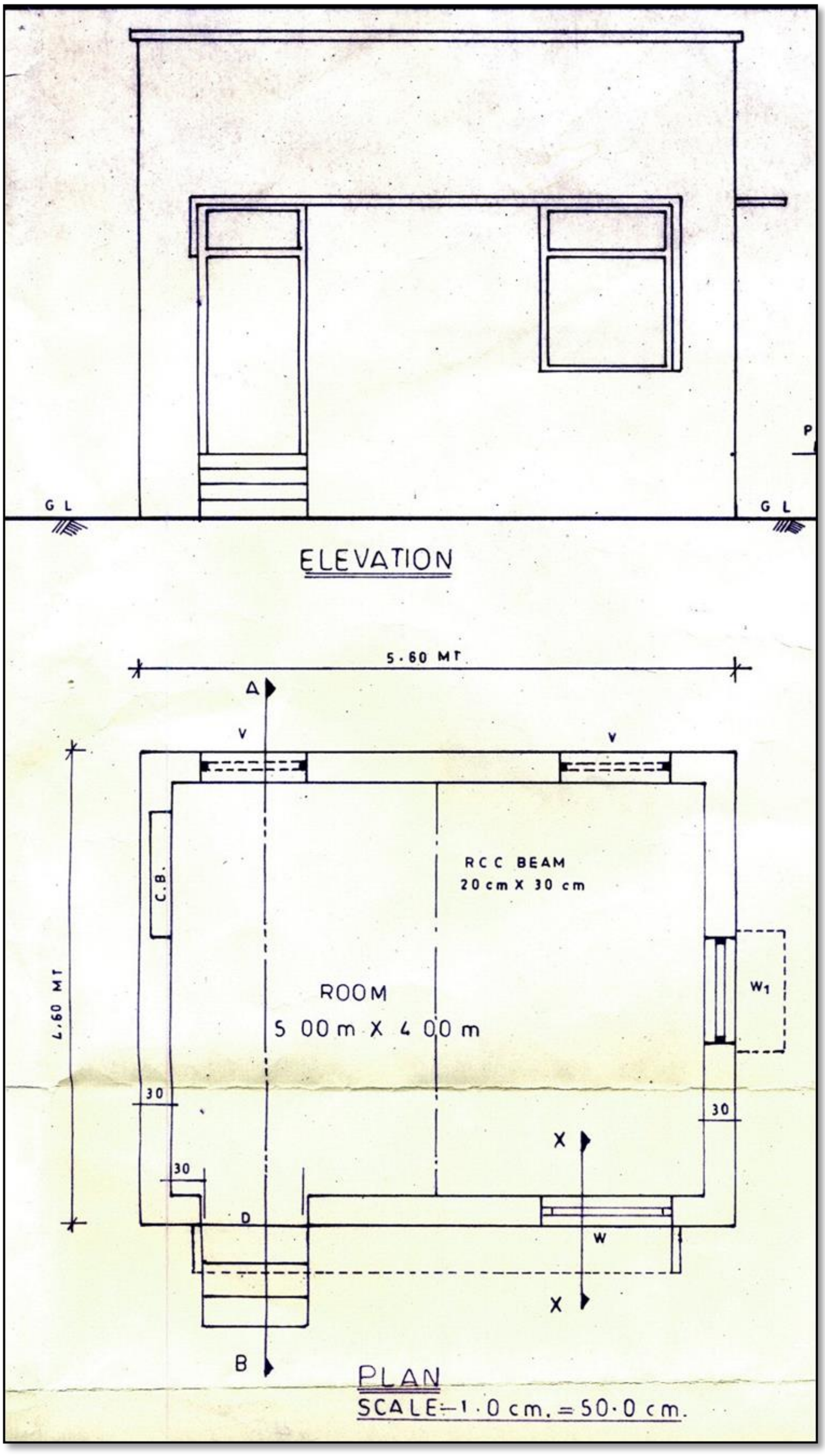
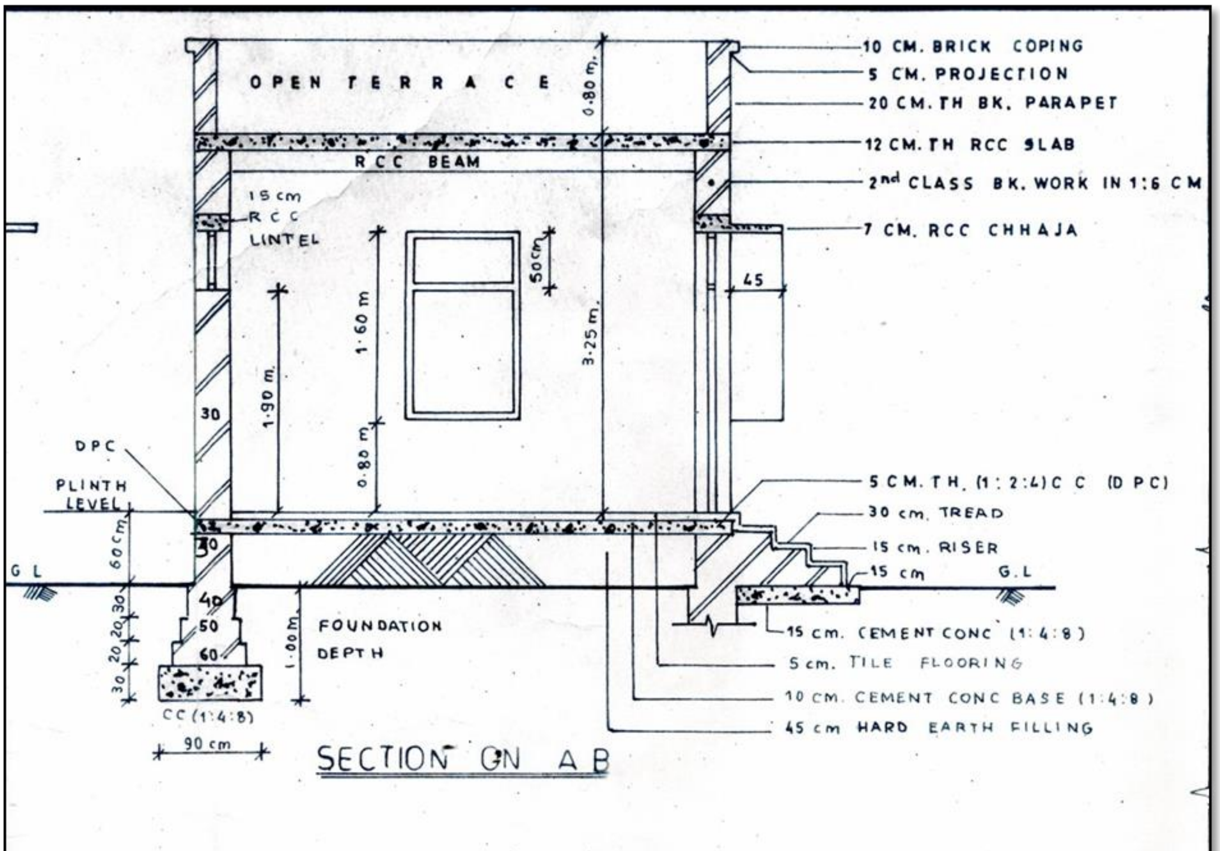


Fig : Plan and Elevation



REFERENCE

Sr no	NAME	CODE	Modular size	SIZE	DESCRIPTION
01	DOOR	D	10 D T 24	1.00 m. X 2.40 m.	FULLY PANELLED
02	WINDOWS	W	12 W T 16	1.20 m. X 1.60 m.	PARTLY GLAZED &
		W1	10 W1 T 16	1.00 m. X 1.60 m.	PARTLY PANELLED
03	VENTILATOR	V	10 V T 05	1.00 m. X 0.50 m.	FULLY GLAZED
04	CUP BOARD	C B	12 CBT 14	1.20 m. X 1.40 m.	WITH TWO SHUTTER

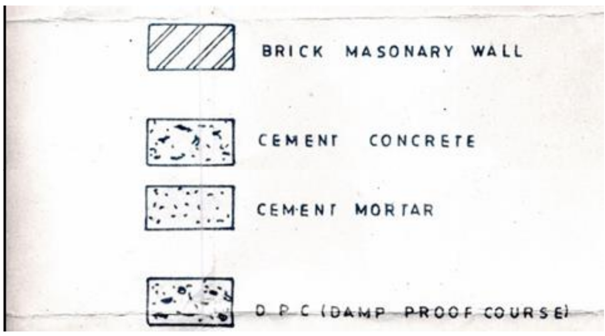


Figure: Section On 'AB' Of A Room

Prepare a drawing of plan elevation and section of the given residential building and sow the schedule of doors and windows.

Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO: 10

ANGULAR MEASUREMENT-COMPASS SURVEY

OBJECTIVE:

To conduct a traverse survey with both surveyor and prismatic compass.

APPARATUS:

Metric chain or Metallic tape, pegs, hammer, plumb bob, ranging rod or ranging pole, Surveyor's compass, Prismatic compass

THEORETICAL BACKGROUND:

Chain surveying can be used when the area to be surveyed is comparatively small and is fairly flat. When large areas are involved in surveying the method of chain surveying is not sufficient and convenient. Hence some other instruments are required for measurements of directions and angles. Direction of the survey line can be established either with relation to each other i.e. angle between two lines or with relation to any other reference line (meridian) i.e. Bearing or inclination.

The instruments used for direct measurement of directions are

1. Surveyor's compass
2. Prismatic compass

The Meridian: The reference line or Meridian is of the following three types

1. True Meridian
2. Magnetic Meridian
3. Arbitrary meridian

Bearing of a line:

Designation of bearing: The common systems of notation of bearings are as shown in figure the angle is measured clockwise from north in the Whole circle bearing system. And that in the Quadrantal bearing system is measured from both North and South in clockwise and anti clockwise direction.

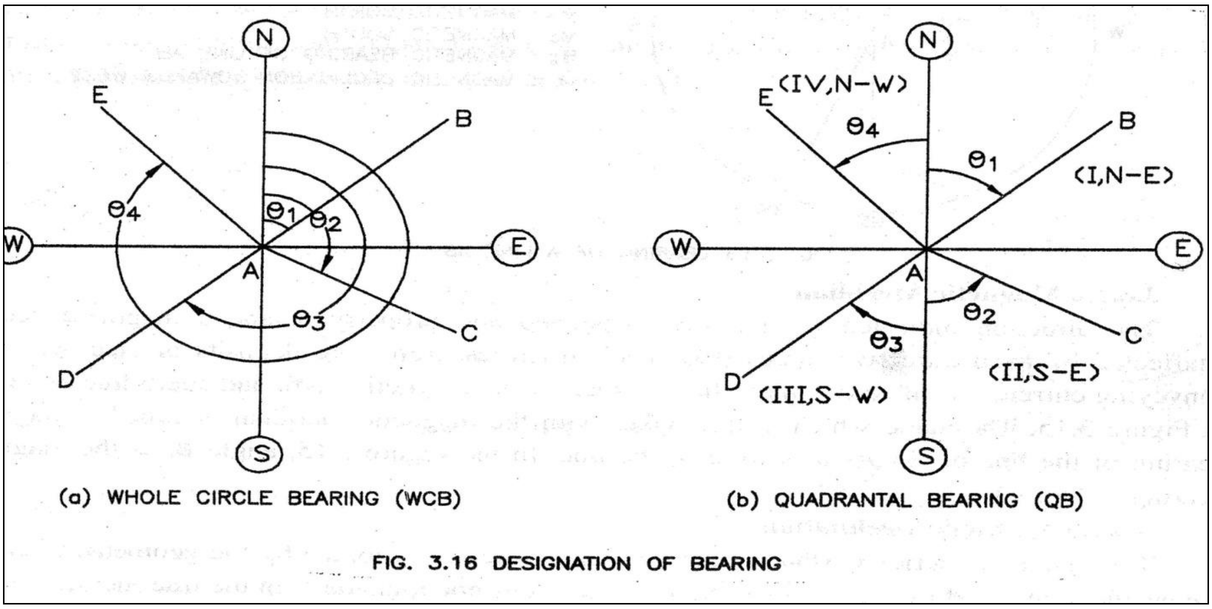


Figure: Designation Of Bearing

The compass:

1. Surveyor's compass: used to give quardrantal bearing of lines.
2. Prismatic compass: used to give whole circle bearing of lines.

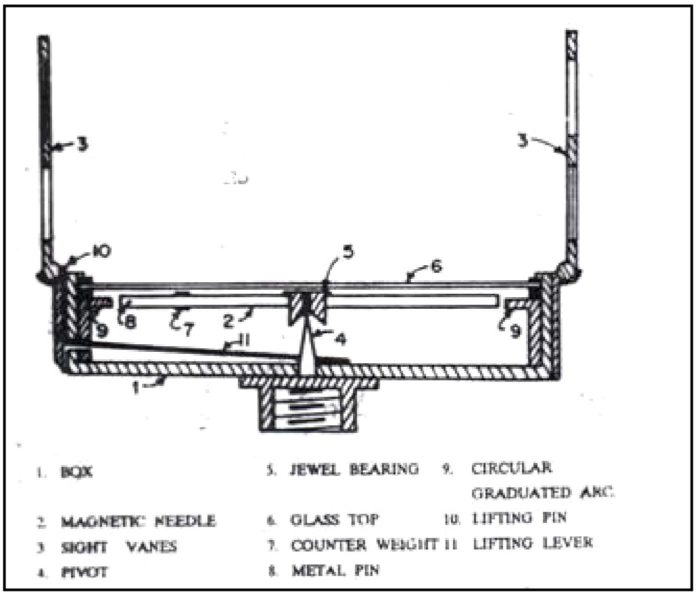


Fig. : Surveyor's compass

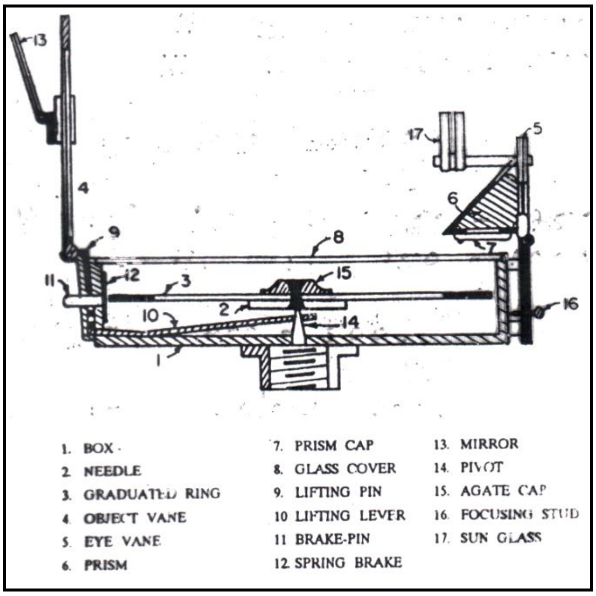


Fig. : Prismatic compass

PROCEDURE:

1. Select the stations on ground to make a close traverse (four stations).
2. Fix the pegs with the help of hammer to mark the stations and nomenclature them in clockwise direction. (ABCD)
3. Measure the distance between successive stations with the help of chain or tape.
4. Affix the Surveyor’s compass on tripod stand and do the temporary adjustments (i.e. Centering and Levelling)
5. Affix two ranging rods at both the adjacent stations of the compass and measure bearings of the survey lines.
6. Note the observed bearings in the observation table.
7. Repeat steps 4 - 6 for all the other stations.

Conduct the survey again with the same procedure by Prismatic Compass.

OBSERVATION TABLE:

Surveyor’s compass

Sr. No.	Line	Length in meter	Fore Bearing	Back Bearing
1.				
2.				
3.				
4.				

Prismatic compass

Sr. No.	Line	Length in meter	Fore Bearing	Back Bearing
1.				
2.				
3.				
4.				

CALCULATION:

1. Plot the traverse with the help of observed bearings.
2. Calculate and observe the difference between fore bearing and back bearing of all the lines if it deviates from 180^0.
3. Calculate all the internal included angles. The sum of included angles in a close traverse must be (2n-4) right angles, if there is any discrepancies in this distribute the error equally to all the angles.
4. Apply necessary corrections for the observed bearings also.
5. Select a suitable scale and plot traverse with the help of corrected bearings and observed lengths.
6. Apply the correction by Bowditch’s rule to correct for the closing error if present.

RESULT TABLES:

Surveyor's compass

Line	Included Angle	Correction	Corrected included angle

Prismatic compass

Line	Included Angle	Correction	Corrected included angle

Marks Obtained:	Signature of Faculty:	Date:

Basics of Civil & Mechanical Engineering (ME-143) **Date:**

EXPERIMENT NO: 11

VERTICAL MEASUREMENT-DUMPY LEVEL

OBJECTIVE:

To measure the reduced levels (R.L.) of the points on ground i.e. measurement in vertical plane.

APPARATUS:

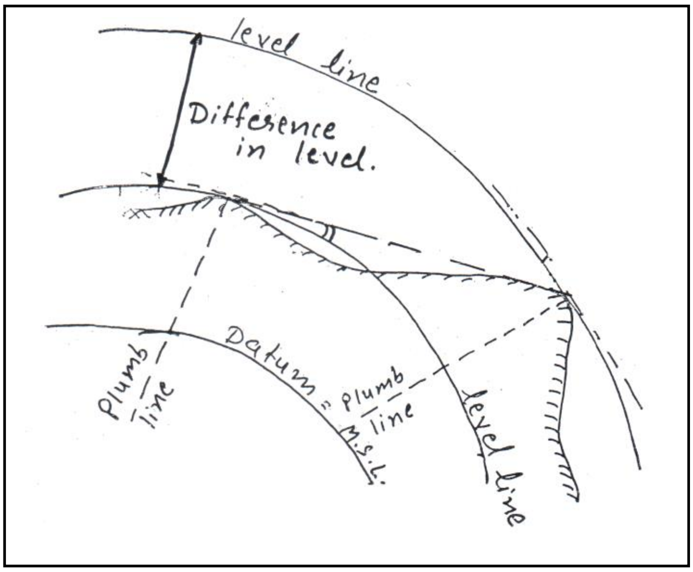
Dumpy Level, Tripod Stand, Leveling Staff, pegs, plumb bob.

THEORETICAL BACKGROUND:

Definitions:

Levelling is an art of determining relative elevation on above or below the surface of earth and setting out points at various heights. The important terminologies in leveling are:

1. Level surface – level line
2. Horizontal line – Vertical Line- Vertical Angle
3. Elevation
4. Mean sea level – reduced level
5. Station



6. Bench mark
    a. GTS bench mark
    b. Permanent bench mark
    c. Temporary bench mark
    d. Arbitrary bench mark
7. Height of instrument
8. Back sight
9. Fore sight
10. Intermediate sight
11. Turning point/ change point

Types of Leveling

1. Direct leveling:

- Simple leveling
- Differential Levelling
- Reciprocal Levelling
- Profile leveling
- Fly leveling
- Check Levelling
- Precise Levelling

2. Indirect leveling :

- Hypsometric leveling
- Trigonometric leveling
- Barometric leveling

DUMPY LEVEL

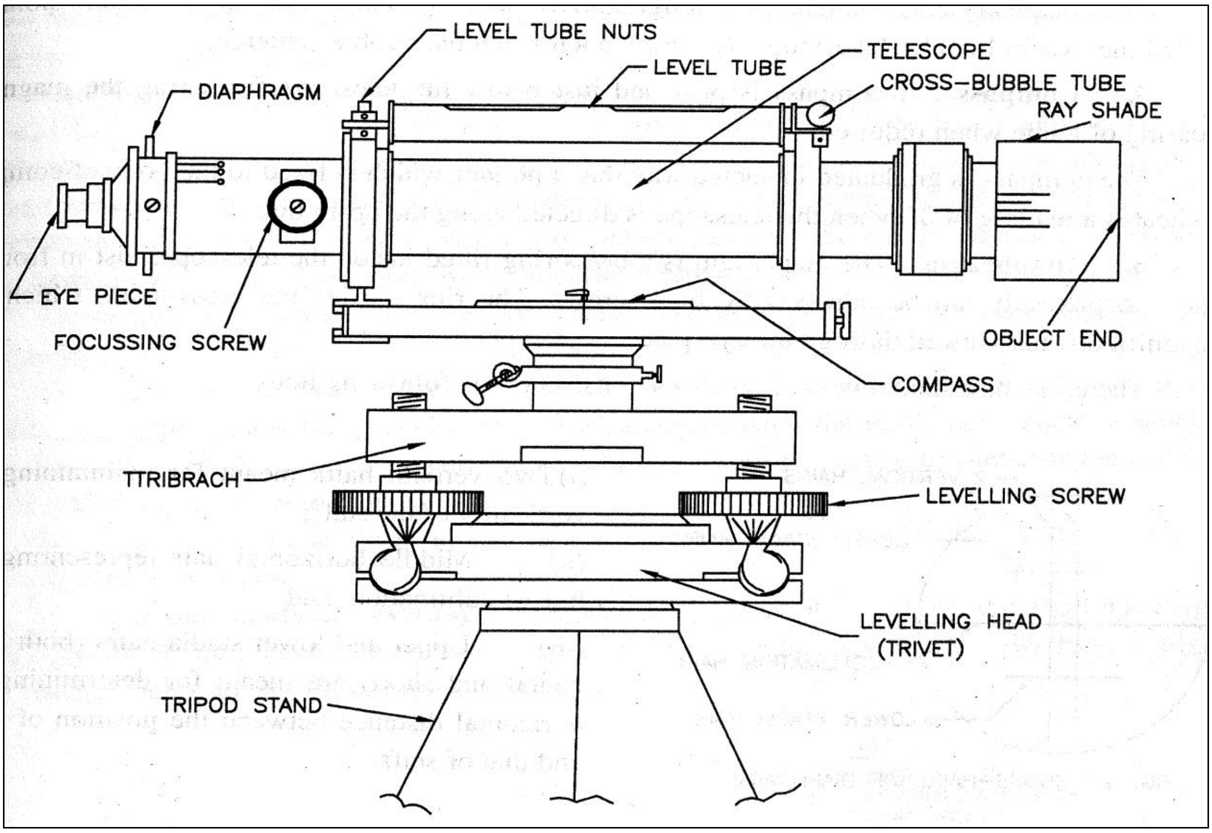


Figure: Dumpy Level

It is used to determine the elevations at different points with respect to the bench mark

LEVELLING STAFF

It is straight rectangular instrument having graduations of surface. It is used for determining the elevation of point above or below the sight.

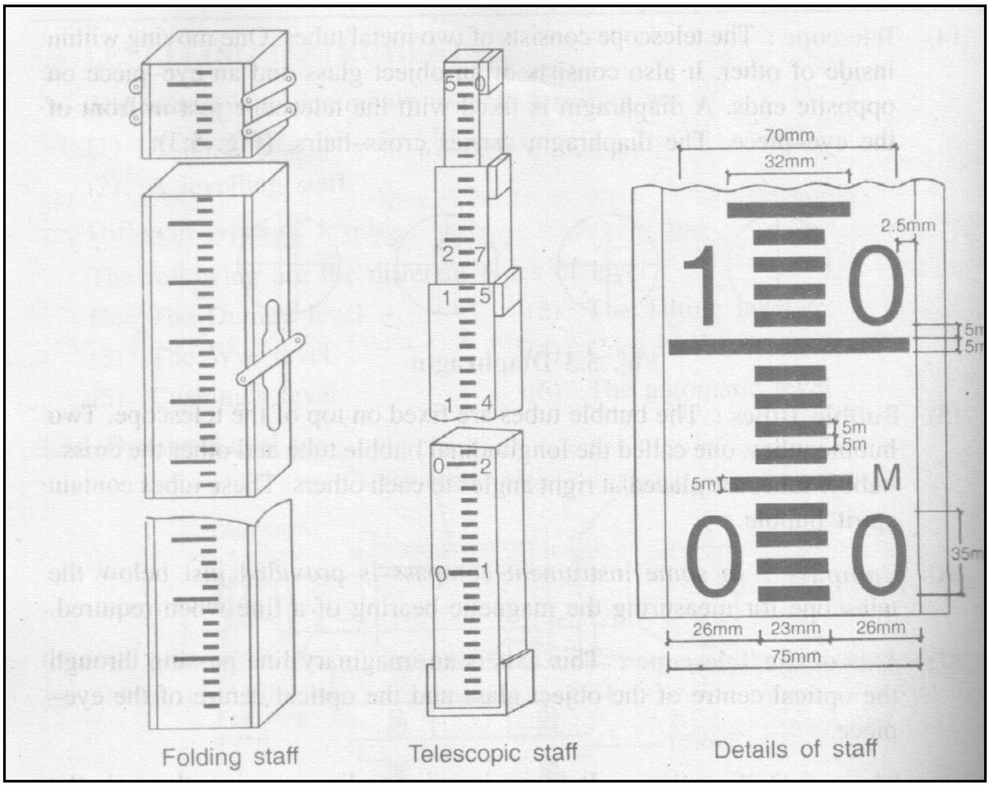


Figure: Leveling staff

PROCEDURE:

1. Mount the dumpy level on the tripod stand.
2. Level the telescope of dumpy level using the three foot screws, so that the line of collimation (axis of telescope) is parallel to the earth's surface.
3. Place the staff on the selected bench mark to measure the back sight.
4. Focus the object glass to the staff and get the clarity of the staff. Take the reading at the bisecting cross wire of the diaphragm of eye piece.
5. Now place the staff on other stations whose elevation is still unknown.
6. Take change point (CP) if required.
7. Record the readings in the observation table.
8. Calculate the reduced levels using Height of Instrument (HI) method and rise & fall method.
9. Check the results using the arithmetic checks for respective tables.

OBSERVATION TABLE:

H.I. METHOD

S. No	STATION	B.S.	I.S.	F.S.	H.I.	RL	Remarks
1	B.M.						
2	1						
3	2						
4	3						
5	4						
6	5						
7	6						
	Arithmetic Check: ∑ B.S. - ∑ F.S. = Last R.L. – First R.L.						

Calculations:

RISE & FALL METHOD

S. No	STATION	B.S.	I.S.	F.S.	RISE	FALL	RL	Remarks
1	B.M.							
2	1							
3	2							
4	3							
5	4							
6	5							
7	6							
	Arithmetic Check: ∑ B.S. - ∑ F.S. = ∑ Rise - ∑ Fall = Last R.L. – First R.L.							

Calculations:

Marks Obtained:	**Signature of Faculty:**	**Date:**